

Koa Shen

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NVIDIA Hardware Hiring Team

Re: Summer 2027 Hardware Engineering Internship

Dear NVIDIA Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, seeking a Summer 2027 hardware internship at NVIDIA. I am particularly interested in mechanical design, electromechanical systems, and hardware validation. NVIDIA's work across accelerated computing and robotics appeals to me because performance depends on physical systems as well as software: careful packaging, manufacturable parts, and hardware that behaves reliably under real operating constraints. My experience connects simulation and mechanical design to fabrication and hands-on testing.

At SLAC National Accelerator Laboratory, I built an open-source Python/Drake digital-twin framework for tightly packaged beamline motion systems. On a 30-DOF, 305-body assembly, millisecond collision checks exposed at least five physical pinch points and informed hardware redesigns. I also redesigned optics mounts, alignment features, and detector strain relief, checked stage and mount load cases around \$100,000-plus detectors, and produced Solid Edge designs with GD&T and PDM-controlled documentation. This work taught me to question model assumptions and turn analysis into concrete design decisions.

With UCSB Gaucho Racing, I led 15 teammates through driver safety-system design, manufacturing, and installation, with the GR26 systems passing FSAE technical inspection. I CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum, holding critical bearing fits within 0.0005 inch. In my own actuator project, I designed a compact 15:1 cycloidal reducer and wrote C++ firmware for stepper control and encoder feedback, bench-validating position correction and missed-step detection on a single joint. These experiences have made me comfortable working across mechanical, software, and fabrication interfaces.

I would bring NVIDIA an engineer who can move between CAD, code, and the shop while keeping tolerances, failure modes, and serviceability in view. I also make work reusable: my SLAC framework includes a 15-stage catalog, and at the Exploratorium I left CAD, CAM files, and remanufacturing instructions so future technicians could maintain my redesigns. I would welcome the opportunity to apply that combination of technical ownership and practical collaboration to NVIDIA's hardware teams.

Thank you for your consideration.

Sincerely,
Koa Shen